Lehrstuhl für Multimediakommunikation
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Team Project

for

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Development of a Structure-from-Motion Pipeline

Entwicklung einer Structure-from-Motion Pipeline

In the field of computer vision, Structure-from-Motion (SfM) is a technique used to reconstruct 3D structures from sequences of 2D images, playing a critical role in applications like photogrammetry, 3D modeling, and autonomous navigation. The goal of this team project is to develop a complete SfM pipeline using open-source software and Python, which can reconstruct 3D structures from fisheye images taken from different viewpoints. For this, the project will use data from a fisheye camera array available at the LMS.

The project begins with a comprehensive literature review, where the team will investigate existing SfM methodologies, focusing on key topics such as feature detection, feature matching, camera pose estimation, and bundle adjustment. Additionally, the team should research fisheye camera calibration and its integration into SfM pipelines. Following the theoretical exploration, the team will gather data by capturing images from various viewpoints using fisheye cameras. Proper calibration of the cameras is essential to ensure accurate reconstruction results.

After collecting the dataset, the team will implement the SfM pipeline. This involves feature detection and matching between images, estimating camera poses, triangulating 3D points, and refining the model through bundle adjustment. Special attention will be given to ensuring the accuracy and robustness of the pipeline under varying conditions such as occlusions or lighting changes. The system will be evaluated using different datasets to assess reconstruction accuracy and computational efficiency.

Once the basic pipeline is in place, the team will extend it to work with data from a fisheye camera array available at the LMS. This step will require adapting the pipeline for the specific fisheye camera properties, particularly through specific calibration techniques and feature detection that account for lens distortions. The ultimate aim is to provide a flexible and modular SfM pipeline that can handle wide-angle camera setups, delivering accurate 3D reconstructions.

The project will be implemented in Python using open-source tools and relevant camera calibration libraries. The final deliverables include a fully functional SfM pipeline with comprehensive documentation, a dataset for testing the pipeline, a detailed evaluation report, and a demonstration of its application to fisheye cameras.

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End: 28.02.2025

(PD Dr.-Ing. J. Seiler)